Database on local labor markets in Mexico: Methodological note

Last updated: April 2024

Content

Database on local labor markets in Mexico: Methodological note

1. Introduction

This document describes the databases that Banco de México publishes for the study of local labor markets in Mexico at the individual, household, and local labor market levels. The datasets, along with the introductory document, data dictionaries, catalogs, and codebook, can be accessed through the following link:

Database (SIDIE Datasets, Banco de México) (banxico.org.mx)

The EconLab is the economic microdata lab at Bank of Mexico. Its main objective is to expand the economic research carried out by this Central Institute, through collaboration with Mexican and foreign researchers.

Any researcher interested in accessing microdata for economic research can contact the EconLab at: econlab@banxico.org.mx

2. Overview & Methodology

The database on local labor markets contains three main sets of files:

- Individual-level microdata
- Household-level microdata
- Aggregated local labor market-level data

The next sections present the general methodology used to construct these datasets.

2.1. Retrieval of historical datasets

The microdata for the Individual and Household datasets were retrieved from INEGI's extended questionnaires for the Population and Housing Censuses of 1990, 2000, 2010, and 2020[1]; as well as the 2015 Intercensal Survey. It is important to highlight that the 2020 census was collected prior to the COVID pandemic, so it does not show the impact of the pandemic. In all cases, the datasets include all variables contained in INEGI's original datasets. The data was downloaded from INEGI's official website: Population and Housing Census (inegi.org.mx).

The aggregated dataset is constructed using microdata on individuals and the aggregation of municipalities into local labor markets.

2.2. Variable harmonization

Variables from different Census waves and the Intercensal Survey are harmonized to make them comparable over time. In the harmonization all existing variables and their differing values across all census years were considered, their catalogs were standardized and, in some cases, values were recalculated from other variables to preserve as many variables as possible.

All the resulting variables are listed in the codebook, where they follow the same order as in the data set. The codebook shows the following fields for each variable:

- variable name,
- description,
- variable type and, for categorical variables, an associated catalog.

All datasets include a variable that identifies the data point's local labor market.

[1] Data for the 2020 Census was collected between March 2 and March 27, 2020, before the Mexican Federal Government declared the start of the health emergency in the country on March 31st. Therefore, these data are not suitable to study the effects of the COVID-19 pandemic.

Section 3 discusses general issues about the harmonization of each dataset, as well as issues relevant to specific variables.

In general, catalogs include a value corresponding to "Not specified", which applies to cases where the information is not available for a given census or intercensal survey wave or the corresponding question was not answered. Therefore, all records have values for all variables that have an associated catalog, and there are no empty or null values. This applies to fields classified with "catalog" or "dimension" column types.

2.3. Star model

To provide more compact microdata sets, these are organized under a star model. This model consists of *dimension* and *facts* files. Dimensions are all variables whose values are defined by a catalog, with each value receiving a unique integer number as a key, and a description. Catalogs for each of the dimensions are stored in a separate file, in the "*Catalogos*" folder. The main *facts* files contain the keys corresponding to the information of each individual or household, along with any continuous-valued variables.

In order to ease the management of granular individual or household-level records in the *facts* files, these files are split by size, resulting in 1 or 2 files per year, which all hold the same variables in the same order.

2.4. Organization of datasets

Due to the large volume of information, the data package consists of folders compressed in.zip format. To access the data, it is necessary to unzip them. The data package includes the following folders:

- **Catalogs** (for microdata)
    - This folder contains a file for each categorical variable, linking the numerical keys assigned by Banco de México to the variable values. In the star model, these correspond to the dimensions.
- **Codes** (only for aggregate data)
    - This folder includes Stata language codes that generate the aggregate database from the microdata files.
- **Data**.
    - **Microdata**: This folder contains the individual data or facts files in a separate .txt or .dta file. For easy handling the data was split into files organized by census wave date.
    - **Aggregates**: This folder contains the aggregate local labor market-level data or facts in a separate .txt or .dta file.

- **Data Dictionary.** This folder contains an Excel file with several sheets detailing the variables, their values in each period, and the decisions made for their harmonization. The first sheet of the Excel file, "*Introducción*", has a brief explanation and the list of variables with links to catalogs, where descriptions, types, and associated catalogs are provided.

  Figures 1 and 2 below illustrate examples of the mentioned folders:

Figure 1. Primary Data File Sheet

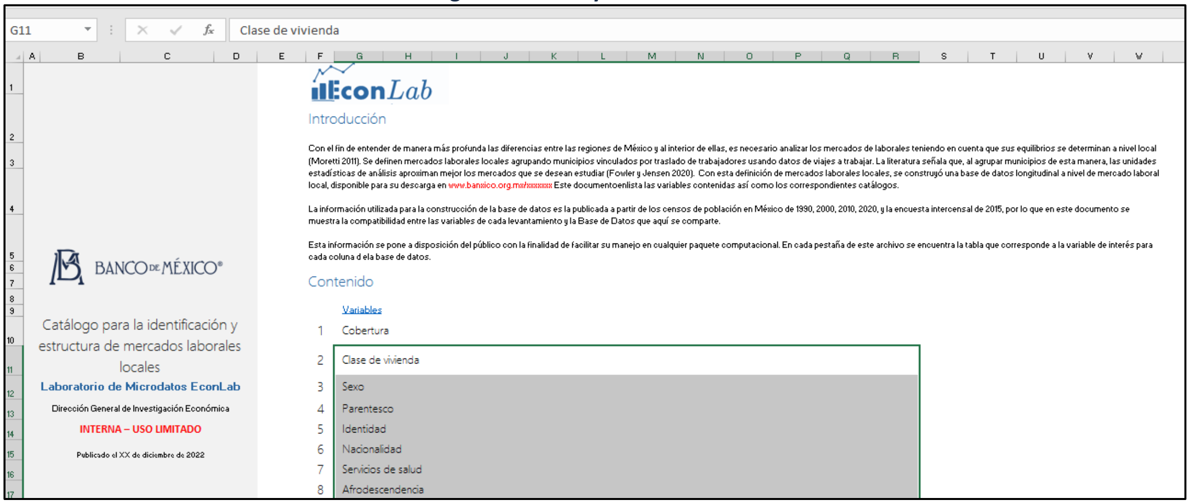
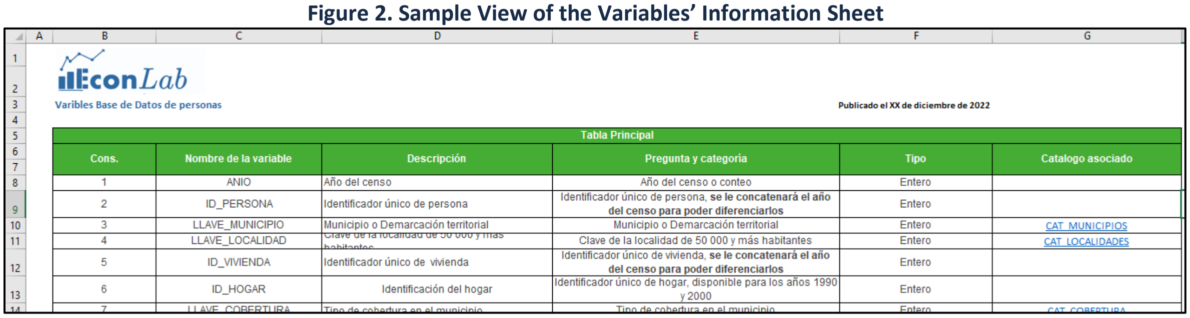


Figure 2. Sample View of the Variables' Information Sheet



3. Specific considerations per dataset

After describing the general aspects, this section provides particular methodological notes for each dataset.

3.1. Individual-level microdata

This dataset has all the information variables from the "*Personas*" questionnaires.

This is with the exception of the information from 1990, for which variables are not divided by whether they correspond to individuals or households. To harmonize these 1990 variables, they were matched to the corresponding variables from other census waves using the descriptions from the questionnaires of other years.

Table 1. Individual-Level Variables

#	Variable name	Description	Variable type
1	ANIO	Census year	Metric/Numeric variable
2	ID_PERSONA	Unique person/individual identifier	Metric/Numeric variable
4	LLAVE_ENTIDAD	State identifier published by INEGI	Catalog/Dimension
5	LLAVE_MUNICIPIO	Municipality or Territorial Demarcation	Catalog/Dimension
6	CLAVE_MUNICIPIO_INEGI	Municipality identifier published by INEGI (AGEM)	Text Fields
7	LLAVE_LOCALIDAD	Key to the town with 50,000 or more inhabitants	Catalog/Dimension
8	CLAVE_LOCALIDAD_INEGI	Location identifier published by INEGI	Text Fields
9	ID_VIVIENDA	Unique household identifier	Metric/Numeric variable
10	LLAVE_COBERTURA	Type of coverage in the municipality	Catalog/Dimension
11	LLAVE_CLASEVIVIENDA	Private housing type	Catalog/Dimension
12	LLAVE_SEXO	Gender	Catalog/Dimension
13	LLAVE_PARENTESCO	Household members relationship	Catalog/Dimension
14	LLAVE_IDENTMADRE	Mother identification	Catalog/Dimension

15	LLAVE_IDENTPADRE	Father identification	Catalog/Dimension
16	LLAVE_PAIS_NAC	Country of birth	Catalog/Dimension
17	LLAVE_ENTIDAD_NAC	Mexican state of birth	Catalog/Dimension
18	LLAVE_NACIONALIDAD	Mexican nationality	Catalog/Dimension
19	LLAVE_SERSALUD	Use of health services	Catalog/Dimension
20	LLAVE_AFRODES	Afro-descendants or Afro-Mexicans	Catalog/Dimension
21	LLAVE_REGISNAC	Birth registry	Catalog/Dimension
22	LLAVE_RELIGION	Religion	Catalog/Dimension
23	LLAVE_HLENGUA	Indigenous language	Catalog/Dimension
24	LLAVE_LENGUAMAT	Indigenous language name	Catalog/Dimension
25	LLAVE_HESPANOL	Spanish speaking ability identification	Catalog/Dimension
26	LLAVE_ELENGUA	Understanding of indigenous language	Catalog/Dimension
27	LLAVE_PERTEINDIGENA	Indigenous self-identification	Catalog/Dimension
28	LLAVE_ASISESCOLAR	School attendance	Catalog/Dimension
29	LLAVE_PAIS_ASISESCOLAR	Country of school attendance	Catalog/Dimension
30	LLAVE_ENTIDAD_ASISESCOLAR	Mexican state of school attendance	Catalog/Dimension
31	LLAVE_MUNICIPIO_ASISESCOLAR	Municipality of school attendance	Catalog/Dimension
32	LLAVE_TIETRASLADO_ESCOLAR	Travel time to school	Catalog/Dimension
33	LLAVE_MEDTRASLADO_ESCOLAR	Mode of transportation to school	Catalog/Dimension
34	LLAVE_NIVACAD	Education (level)	Catalog/Dimension
35	LLAVE_CARRERA	Higher education field name	Catalog/Dimension
36	LLAVE_ALFABETISMO	Literacy	Catalog/Dimension
37	LLAVE_PAIS_RES5A	Country of residence five years ago	Catalog/Dimension
38	LLAVE_ENTIDAD_RES5A	Mexican state of residence five years ago	Catalog/Dimension
39	LLAVE_MUNICIPIO_RES5A	Mexican municipality of residence five years ago	Catalog/Dimension
40	LLAVE_CAUSAMIGRACION	Cause of migration	Catalog/Dimension
41	LLAVE_SITUACONYUGAL	Marital status	Catalog/Dimension
42	LLAVE_IDENTPAREJA	Partner identification	Catalog/Dimension
43	LLAVE_ACTPRIMARIA	Current workforce participation	Catalog/Dimension
44	LLAVE_OCUPACION	Occupation name	Catalog/Dimension
45	LLAVE_SITTRA	Job position	Catalog/Dimension
46	LLAVE_ACTECONOMICA	Activity of the business, company or workplace	Catalog/Dimension
47	ACTIVIDAD_ECONOMICA_INEGI	Economic activity identifier published by INEGI, this code corresponds to SCIAN as of 2000.	Text Fields
48	LLAVE_PAIS_TRABAJO	Country of work	Catalog/Dimension
49	LLAVE_ENTIDAD_TRABAJO	Mexican state of work	Catalog/Dimension
50	LLAVE_MUNICIPIO_TRABAJO	Mexican municipality where the workplace is located	Catalog/Dimension
51	LLAVE_TIETRASLADO_TRABAJO	Commuting time to work	Catalog/Dimension
52	LLAVE_MEDTRASLADO_TRABAJO	Transportation means to work	Catalog/Dimension
53	LLAVE_TAMLOC	Locality size	Catalog/Dimension
54	ESTRATO	Strata	Text Fields
55	UPM	Primary sampling unit	Text Fields
56	FACTOR_EXP	Expansion Factor	Metric/Numeric variable
57	NUMPERSONA	Person/individual number	Metric/Numeric variable
58	EDAD	Age	Metric/Numeric variable
59	ESCOLARIDAD	Education level (grade)	Metric/Numeric variable
60	ESCOLARIDAD_ACUMULADA	Total educational attainment (years)	Metric/Numeric variable
61	INGRESO	Monthly employment income	Metric/Numeric variable
62	HORAS_TRABAJADAS	Hours worked last week	Metric/Numeric variable
63	HIJOS_NACIDOS	Number of children born in the family	Metric/Numeric variable
64	HIJOS_VIVOS	Number of living children in the family	Metric/Numeric variable
65	HIJOS_FALLECIDOS	Number of deceased children in the family	Metric/Numeric variable
66	MERCADO_TRABAJO_LABORAL	Local labor market identifier	Metric/Numeric variable

3.1.1. Particular considerations

1) PERSON_ID

For this variable, in the 1990, 2000, and 2010 censuses, the "ID_PERSONA" variable as defined by INEGI was either unavailable or not unique at the national level. Thus, for these years a sequential number was assigned, after sorting the records by entity, municipality, household identifier, and individual number.

For the years 2015 and 2020, the "ID_PERSONA" variable provided by the census was retained. Additionally, the census year was added to all records in the first 4 digits of this variable.

2) INCOME

For this variable, the value "999998" in the 2000, 2010, 2015, 2020 censuses corresponds to an income of 999999 pesos per month or more. In the 1990 wave income data is not censored in this way and so the value "999998" represents a measured level of income. For this reason the value representing top-censored income was replaced with the value "99999999".

3) Other variables

For information on variables that may have more than one value, additional structures were created. These are the variables referring to disabilities, entitlement, income and benefits and the details can be found below.

Disabilities

Table 2. Variables Identifying Disabilities in the Individual-Level Dataset

#	Variable name	Description
1	ANIO	Census year
2	ID_PERSONA	Unique person/individual identifier
3	LLAVE_DISCAPACIDAD	Disability identifier
4	LLAVE_CAUSADISC	Cause of Disability

Employment benefits

Table 3. Variables Identifying Employment Benefits in the Individual-Level Dataset

#	Variable name	Description
1	ANIO	Census year
2	ID_PERSONA	Unique person/ individual identifier
3	LLAVE_PRESTACION	Employment benefits

Income

Table 4. Variables Identifying Income Information in the Individual-Level Dataset

#	Variable name	Description
1	ANIO	Census year
2	ID_PERSONA	Unique person/individual identifier
3	LLAVE_INGRESO	Type of income identifier Note: Due to a change in the methodology of the census of persons, which does not specify to which person in the household the type of income corresponds, the detail of income for the years 2015 and 2020 was assigned to the head of household of each household.
4	INGRESO	Income

Entitlement

Table 5. Variables Identifying Health Service Entitlements in the Individual-Level Dataset

#	Variable name	Description
1	ANIO	Census year
2	ID_PERSONA	Unique person/individual identifier
3	LLAVE_DHSERSAL	Health services affiliation

3.2. Household-level microdata

3.2.1. Variable structure, data at the household level

Table 6. Household-Level Variables

#	Variable name	Description	Column type
1	ANIO	Census year	Metric/Numeric variable
2	ID_VIVIENDA	Dwelling identifier	Metric/Numeric variable
3	ID_HOGAR	Household identifier	Metric/Numeric variable
4	LLAVE_ENTIDAD	State identifier published by INEGI	Catalog/Dimension
5	LLAVE_MUNICIPIO	Municipality	Catalog/Dimension
6	CLAVE_MUNICIPIO_INEGI(CLAVE_DE_AGEM)	Municipality identifier published by INEGI (AGEM)	Text Fields
7	LLAVE_LOCALIDAD	Location	Catalog/Dimension
8	CLAVE_LOCALIDAD_INEGI	Location identifier published by INEGI	Text Fields
9	LLAVE_TAMLOC	Location size	Catalog/Dimension
10	LLAVE_VIVIENDAPART	Classification of private housing.	Catalog/Dimension
11	LLAVE_MATERIALPAREDES	Type of wall material	Catalog/Dimension
12	LLAVE_MATERIALTECHOS	Type of roof material	Catalog/Dimension
13	LLAVE_MATERIALPISOS	Type of flooring material	Catalog/Dimension
14	LLAVE_COCINADISPONIBLE	Availability of kitchen	Catalog/Dimension
15	LLAVE_LUGARDONDECOCINAN	Availability of any cooking area	Catalog/Dimension
16	LLAVE_COCINADORMITORIO	Kitchen bedroom	Catalog/Dimension
17	LLAVE_SANITARIODISPONIBLE	Availability of toilet	Catalog/Dimension
18	LLAVE_SANITARIOEXCLUSIVO	Exclusive use of the toilet	Catalog/Dimension
19	LLAVE_SANITARIOAGUA	Toilet with water connection	Catalog/Dimension
20	LLAVE_DOTACIONAGUA	Water provision frequency	Catalog/Dimension
21	LLAVE_ABASTOAGUA	Water supply source	Catalog/Dimension
22	LLAVE_AGUAPOTABLE	Piped water	Catalog/Dimension
23	LLAVE_AGUANOPOTABLE	Non-piped water	Catalog/Dimension
24	LLAVE_DRENAJE	Sewer system	Catalog/Dimension
25	LLAVE_ELECTRICIDAD	Electricity	Catalog/Dimension
26	LLAVE_COMBUSTIBLE	Type of Cooking Fuel	Catalog/Dimension
27	LLAVE_VIVIENDATENENCIA	Housing Tenure	Catalog/Dimension
28	LLAVE_VIVIENDAADQUISICION	Housing ownership status at the time of purchase	Catalog/Dimension
29	LLAVE_TIPOHOGAR	Household type	Catalog/Dimension
30	LLAVE_BASURAELIMINACION	Waste disposal	Catalog/Dimension
31	LLAVE_CONSTRUCCION	Building age	Catalog/Dimension
32	NFOCOS	Total number of light bulbs in the house	Metric/Numeric variable
33	NFOCOSAHORRADORES	Total number of energy-saving light bulbs in the house	Metric/Numeric variable

34	TOTALCUARTOS	Total number of rooms	Metric/Numeric variable
35	NCUARTOSDORMIR	Number of bedrooms	Metric/Numeric variable
36	NPERSONASVIVIENDA	Number of people in the dwelling	Metric/Numeric variable
37	NFAMILIASVIVIENDA	Number of families or groups in the dwelling	Metric/Numeric variable
38	NUMHOGAR	Household sequence number in the dwelling	Metric/Numeric variable
39	ESTRATO	Strata	Text Fields
40	UPM	Primary sampling unit	Text Fields
41	FACTOR_EXP	Expansion Factor	Metric/Numeric variable
42	MERCADO_TRABAJO_LOCAL	Local labor market identifier	Metric/Numeric variable

3.2.2. Particular considerations

The 1990 and 2000 waves of the Census recorded each household as a separate observation, even if they lived in the same dwelling. For this reason, the "ID_VIVIENDA" (corresponding to the dwelling identifier) was not unique at the national level. For these years a sequential household number was assigned after ordering the records by the following variables:

- 1990
  - Entity, Municipality, size of locality, type of private home, number of people, walls, total number of rooms and housing folio.
- 2000
  - Entity, Municipality, locality, home number and housing number

From 2010 on, INEGI treats all individuals that live in the same dwelling as a single household. For this reason, the data presented here preserve the values of "ID_VIVIENDA" from the Census for the years 2010, 2015 and 2020 as household identifiers.[2] Additionally, the census year was included in the first 4 digits of all records in this variable.

3.2.3. Other variables

As in the individual section, variables for which it is possible to have more than one value were split, in the case of this dataset these are: equipment, waste separation and assets. The tables for these structures are provided below:

Assets

Table 7. Variables identifying asset possessions in the household disaggregated dataset

#	Variable name	Description	Type
1	ANIO	Census year	Metric/Numeric variable
2	ID_VIVIENDA	Household identification	Metric/Numeric variable
3	LLAVE_BIEN	Describe the assets the household possesses.	Catalog/Dimension

Equipment

[2] In the original data from the 2010 Census, "ID_VIVIENDA" identifies households within each state, and not at the national level. For this reason, the "ID_VIVIENDA" field presented here is the unique sequential number corresponding to each observation after sorting the database by state identifier and the household ID in the original data.

Table 8. Variables identifying equipment possessions in the household disaggregated dataset

#	Variable name	Description	Type
1	ANIO	Census year	Metric/Numeric variable
2	ID_VIVIENDA	Household identification	Metric/Numeric variable
3	LLAVE_EQUIPAMIENTO	Describe the equipment the household possesses	Catalog/ Dimension

Waste separation

Table 9. Variables for identifying waste separation practices in the household disaggregated dataset

#	Variable name	Description	Type
1	ANIO	Census year	Metric/Numeric variable
2	ID_VIVIENDA	Household identification	Metric/Numeric variable
3	LLAVE_SEPARACIONBASURA	Describe the household's method for separating waste.	Catalog/ Dimension

3.3. Local labor market level aggregates

As a third dataset, aggregates were created at the level of local labor markets with information on the variables of individuals including demographic data, education, economic sectors, income, type of employment, etc.

The details of these local labor market-level variables can be found in the data dictionary. The data sets generated at the aggregate level are averages or sums at the local labor market level. The data files for these variables are organized by topic and type.

For the poverty indicators in the dataset named "*Vulnerabilidad*", the fraction of households with incomes below the poverty lines defined by Coneval is calculated by aggregating the individual incomes of all members and comparing the result to the household size-adjusted poverty line, obtained by multiplying the individual poverty line by the household size.

The Stata code used to produce these local labor market aggregates is available in the "*Código*" folder. The codes presented generate each of the data files described below:

Table 10. List of Local Labor Market Aggregates Files.

Cons.	Folder	Filename	Description
0	Vulnerabilidad	Vulnerabilidad.dta	Dataset at the local labor market-year level, including social vulnerability indicators such as poverty rates, child labor, minors with a single parent, etc.
1	Informalidad	Informalidad.dta	Dataset at the local labor market-year level, including measures of informal employment.
2	Demográficos	Demograficos_Nivel.dta	Dataset at the local labor market-year level, with demographic and employment variables and ratios (i.e proportion of the working-age population), expressed in absolute terms.
3	Demográficos	DemograficosLogs.dta	Dataset at the local labor market-year level, with demographic and employment variables, and ratios (i.e proportion of the working-age population), expressed in logarithmic terms.

4	Ingresos_Pob15	Ingresos_Pob15s.dta	Dataset at the individual-year level with various income measures. It is used to generate metrics of residual/adjusted income.
5	Salarios Residuales	SalResMTL_N.dta	Dataset at the local labor market–year level, with residual average income variables obtained after controlling for education and age, for the entire population.
6	Salarios Residuales	SalResMTL_0.dta	Dataset at the local labor market–year level, with residual average income variables obtained after controlling for education and age, for women.
7	Salarios Residuales	SalResMTL_1.dta	Dataset at the local labor market–year level, with residual average income variables obtained after controlling for education and age, for men.
8	Salarios Residuales	Convergencia_Nivel_MTL_N.dta	Year-level dataset with measures of inequality and convergence between local labor markets. It includes various metrics of convergence, such as standard deviation of income (between local labor markets), income percentiles, and percentile ratios. All variables are calculated in a sample of men and women.
9	Salarios Residuales	Convergencia_Nivel_MTL_0.dta	Year-level dataset with measures of inequality and convergence between local labor markets. It includes various metrics of convergence, such as standard deviation of income (between local labor markets), income percentiles, and percentile ratios. All variables are calculated in a sample of women.
10	Salarios Residuales	Convergencia_Nivel_MTL_1.dta	Year-level dataset with measures of inequality and convergence between local labor markets. It includes various metrics of convergence, such as standard deviation of income (between local labor markets), income percentiles, and percentile ratios. All variables are calculated in a sample of men
11	Salarios Residuales	SalariosResiduales.dta	Dataset at the individual-year level with income variables residualized by education and age.
12	Choques Bartik	LongBartikNacional.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated across all industries and the entire employed population.
13	Choques Bartik	LongBartikHombre0.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated across all industries and only women's employment for national changes.
14	Choques Bartik	LongBartikHombre1.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes).

			Calculated across all industries and only men's employment for national changes.
15	Choques Bartik	WideBartikNacional.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes), as well as a variable for each industry representing the share of employment in 1990 in the local labor market and percentage growth in total national employment.
16	Choques Bartik	WideBartikHombre0.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes), as well as a variable for each industry representing the share of employment in 1990 in the local labor market and percentage growth in national female employment.
17	Choques Bartik	WideBartikHombre1.dta.	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes), as well as a variable for each industry representing the share of employment in 1990 in the local labor market and percentage growth in national male employment.
18	Choques Bartik	LongBartikNacional_Trade0.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across non-tradable industries and the entire employed population for national changes.
19	Choques Bartik	LongBartikNacional_Trade1.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across tradable industries and the entire employed population for national changes.
20	Choques Bartik	LongBartikNacional_Manuf.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across manufacturing industries and the entire employed population for national changes.
21	Choques Bartik	LongBartikHombre0_Trade0.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across non-tradable industries and female employment for national changes.
22	Choques Bartik	LongBartikHombre0_Trade1.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across tradable industries and the entire employed population for national changes.

23	Choques Bartik	LongBartikHombre0_Manuf.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across manufacturing industries and female employment for national changes.
24	Choques Bartik	LongBartikHombre1_Trade0.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across non-tradable industries and male employment for national changes.
25	Choques Bartik	LongBartikHombre1_Trade1.dta	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across tradable industries and male employment for national changes.
26	Choques Bartik	LongBartikHombre1_Manuf	Dataset at the local labor market-year level, with Bartik shock variables (in two forms: using employment and wages for national changes). Calculated only across manufacturing industries and male employment for national changes.
27	Choques Bartik	Tasa_Crecimiento_Industria_Nacional.dta	Dataset at the industry-year level, with various metrics of change in wages and employment in consecutive periods, for total employment.
28	Choques Bartik	Tasa_Crecimiento_Industria_Hombre0.dta	Dataset at the industry-year level, with various metrics of change in wages and employment in consecutive periods, for female employment.